

Task 4: Insurance Claim Prediction Report

Introduction

The objective of this project is to analyze a medical insurance dataset and develop a machine learning model capable of predicting insurance charges. Insurance companies determine medical costs based on several factors, including age, body mass index (BMI), smoking habits, number of children, and geographical region. By exploring the dataset and applying machine learning techniques, this project demonstrates how data analytics can be used to estimate future insurance expenses and identify the factors that have the greatest impact on medical costs.

Dataset Understanding

The Medical Cost Personal Dataset contains 1,338 records and 7 attributes. The dataset includes demographic and health-related information about individuals along with their medical insurance charges.

Dataset Features

- **Age:** Age of the individual.
- **Sex:** Gender of the individual.
- **BMI:** Body Mass Index, a measure of body fat based on height and weight.
- **Children:** Number of dependents covered by insurance.
- **Smoker:** Indicates whether the individual is a smoker or non-smoker.
- **Region:** Residential region of the individual.
- **Charges:** Medical insurance cost charged to the individual.

The dataset was examined using dataset shape, column information, sample records, and summary statistics. No missing values were found, making the dataset suitable for analysis and machine learning model development.

Visualizations

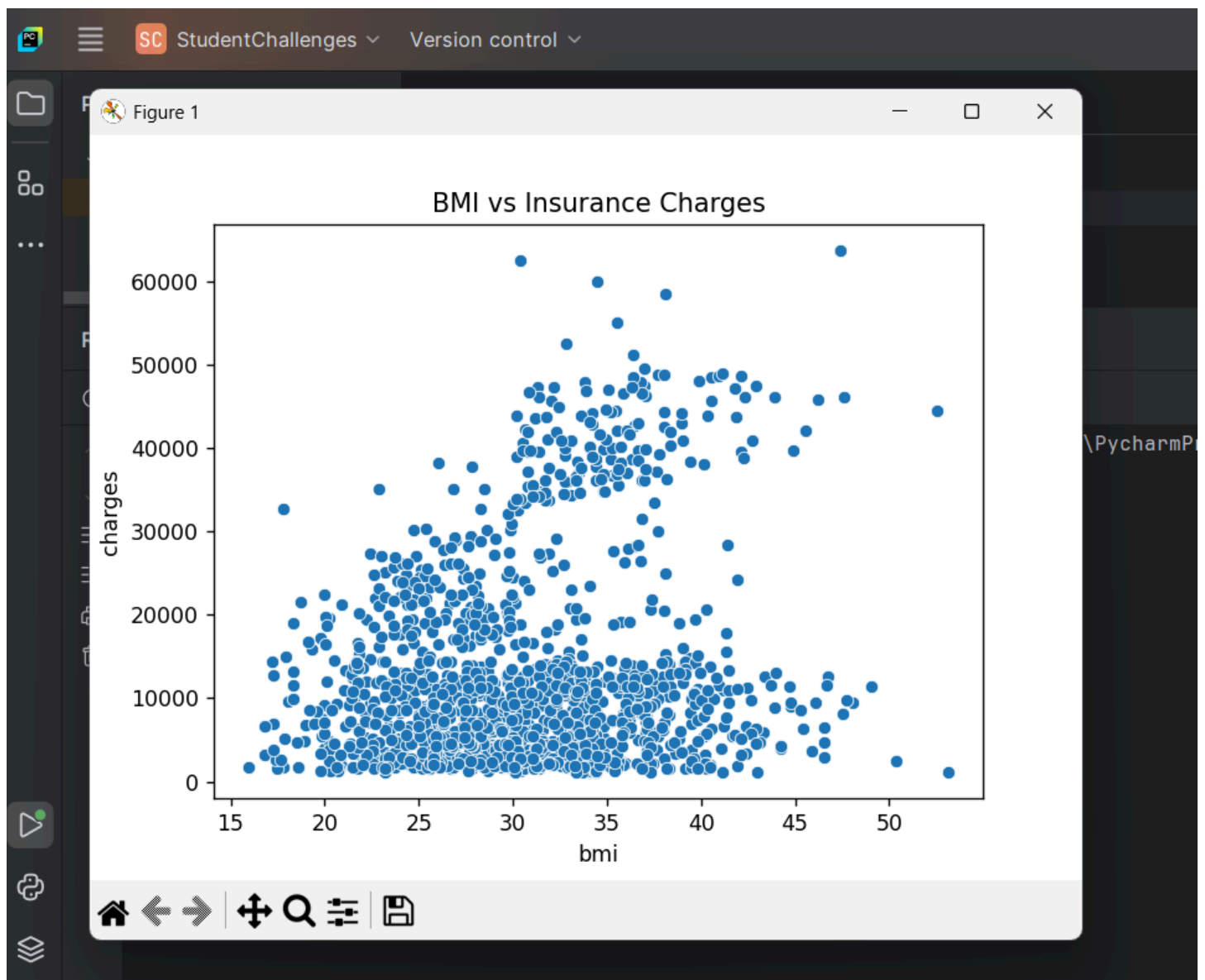


Figure 1: BMI vs Insurance Charges Observation: There is a moderate positive relationship between BMI and insurance charges

Observation

The scatter plot shows a moderate positive relationship between BMI and insurance charges. Individuals with higher BMI values generally tend to incur higher medical costs. However, the spread of the data points suggests that BMI alone does not determine insurance charges, and other factors also contribute significantly.

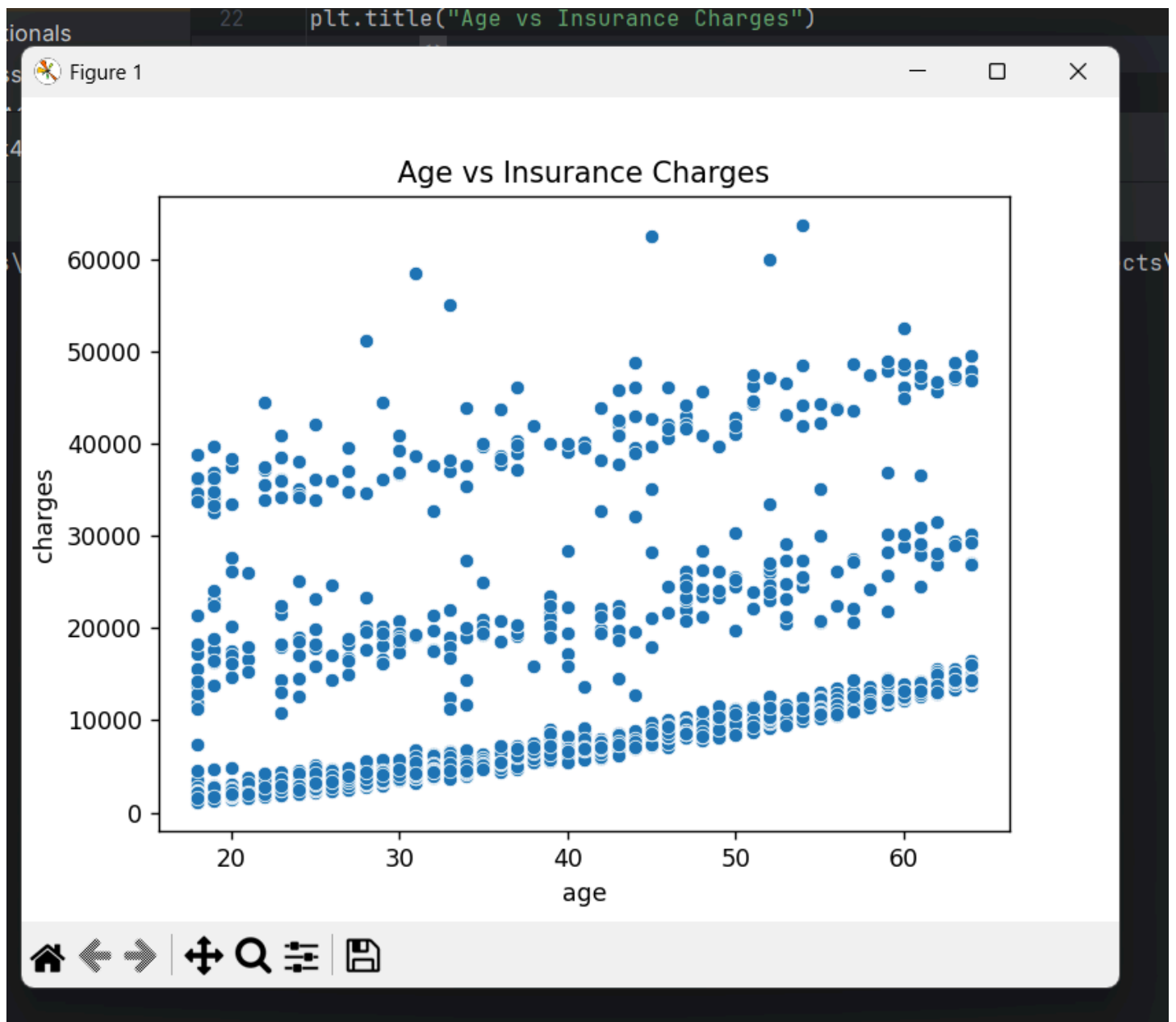


Figure 2: Age vs Insurance ChargesObservation: Insurance charges generally increase with age.

Observation

The visualization indicates that insurance charges generally increase with age. Older individuals tend to have higher medical expenses compared to younger individuals. This suggests that age is an important predictor of insurance costs and should significantly influence the model's predictions.

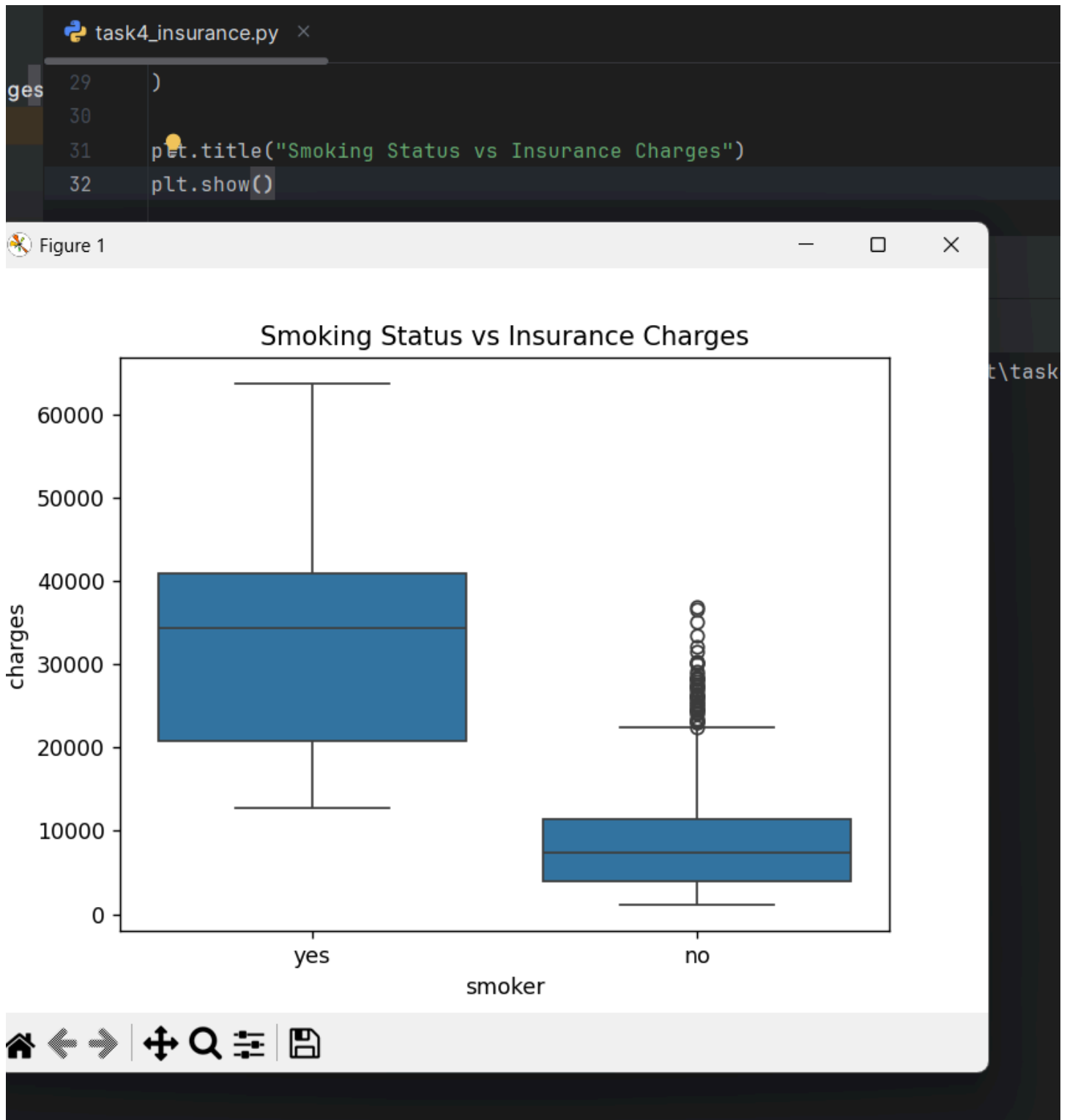


Figure 3: Smoking Status vs Insurance ChargesObservation: Smokers have significantly higher insurance charges than non-smokers.

Observation

The box plot clearly shows that smokers have significantly higher insurance charges than non-smokers. The median insurance cost for smokers is substantially greater, indicating that smoking is one of the strongest factors affecting medical insurance expenses.

Data Preprocessing

Before training the machine learning model, categorical variables such as sex, smoker status, and region were converted into numerical form using one-hot encoding. This step allows the Linear Regression algorithm to process categorical information effectively.

The target variable selected for prediction was insurance charges, while all other attributes were used as input features. The dataset was then divided into training and testing sets using an 80:20 ratio to evaluate model performance on unseen data.

Model Development

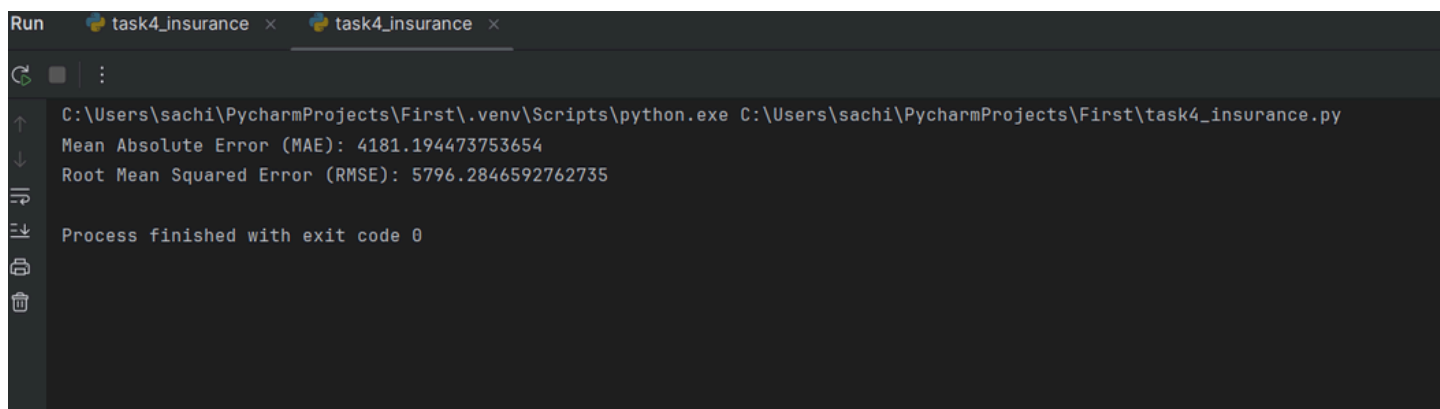
A Linear Regression model was chosen because the target variable, insurance charges, is continuous in nature. Linear Regression attempts to establish a mathematical relationship between the independent variables and the target variable to predict future values.

The model was trained using the training dataset and then evaluated on the testing dataset to measure its predictive performance.

Model Building

Linear Regression was used. Categorical variables were encoded, the dataset was split into training and testing sets, and the model was trained on the training data.

Model Evaluation



```
Run task4_insurance x task4_insurance x
C:\Users\sachi\PycharmProjects\First\.venv\Scripts\python.exe C:\Users\sachi\PycharmProjects\First\task4_insurance.py
Mean Absolute Error (MAE): 4181.194473753654
Root Mean Squared Error (RMSE): 5796.2846592762735
Process finished with exit code 0
```

Mean Absolute Error (MAE): 4181.19Root Mean Squared Error (RMSE): 5796.28Observation: The model predicts insurance charges with reasonable accuracy. Smoking status, age, and BMI are important factors affecting insurance costs.

Conclusion

This project successfully analyzed the Medical Cost Personal Dataset and developed a Linear Regression model to predict insurance charges. Exploratory Data Analysis revealed that age, BMI, and smoking status have significant effects on insurance costs. Among all variables, smoking status emerged as one of the strongest predictors of insurance charges.

The Linear Regression model achieved a Mean Absolute Error of 4181.19 and a Root Mean Squared Error of 5796.28, indicating acceptable predictive performance. Overall, the project demonstrates how data analytics and machine learning techniques can be applied to solve real-world problems in the insurance industry and support data-driven decision-making.